

International Institute of Information Technology, Hyderabad
(Deemed to be University)

Statistical Methods in AI (CS7.403) - Monsoon-2025

Quiz 1 (Set A)

Maximum Time : 45 Minutes

Total Marks : 35

Roll No. _____

Seat No. _____

Date. 29-08-2025

Invigilator Sign. _____

Marks secured

Question:	1	2	3	4	5	Total
Points:	6	8	10	6	5	35
Score:						

General Instructions to the students

1. Answers in Rough Work will not be graded.
2. Do not answer with pencil.
3. **No questions will be answered during the exam. Make necessary reasonable assumptions, state them and proceed.**
4. If any student is found indulging in malpractice or copying in the examination hall, the student will be given 'F' grade for the course and may be debarred from writing other examinations.

Questions

1. (6 points) For each of the following variable, what is (i) its type (e.g. 'qualitative', 'quantitative', 'ordinal' etc.). Provide the most detailed answer, e.g. "qualitative → ordinal" (ii) a suitable method for encoding the variable. Briefly explain your reasoning for part (ii).
(a) Pin code as input variable

(b) Roll number as input variable

(c) Input variable is the angle made by smartphone's flat surface with respect to a vertical (gravity) axis.

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2. (8 points) Given two vectors $x = \{x_1, x_2, \dots, x_N\}$ and $y = \{y_1, y_2, \dots, y_N\}$, with sample means $\bar{x} = \frac{1}{N} \sum_{i=1}^N x_i$, $\bar{y} = \frac{1}{N} \sum_{i=1}^N y_i$, and sample standard deviations denoted by $\sigma_x = \sqrt{\frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})^2}$, $\sigma_y = \sqrt{\frac{1}{N-1} \sum_{i=1}^N (y_i - \bar{y})^2}$, respectively.
- (a) (3 points) Write the formula for correlation coefficient ρ between x and y in terms of the above quantities. Assume that feature values are scalars.

(b) (5 points) Prove that $|\rho| \leq 1$.

- Cauchy-Schwarz : $|\langle \mathbf{a}, \mathbf{b} \rangle| \leq \|\mathbf{a}\| \|\mathbf{b}\|$
- $A = \sqrt{\sum_i a_i^2}$, $B = \sqrt{\sum_i b_i^2}$ and $p_i = \frac{a_i}{A}$, $q_i = \frac{b_i}{B}$. Use $(p_i \pm q_i) \geq 0$
- Let $a = \sum_i (x_i - \bar{x})$ and $b = \sum_i (y_i - \bar{y})$. Use $(a - sb)^2 \geq 0 \Rightarrow$ discriminant $D \leq 0$

3. (10 points) There are 7 test samples and 4 classes (A, B, C, D). The classifier is k-nearest neighbor. For each sample, the ground-truth class is given in the first column in square brackets. The distances of the 3 nearest neighbors from training set and corresponding class labels are given in the table below. Answer the questions below. For results involving fractions, you can leave it as the final simplified fraction - exact decimal calculation is not required. If there is no majority class in the k nearest neighbors, first nearest neighbor's label is chosen as the prediction.

Test Sample	Neighbor 1 (Class, Distance)	Neighbor 2 (Class, Distance)	Neighbor 3 (Class, Distance)
1 [B]	(A, 0.2)	(B, 0.3)	(A, 0.4)
2 [B]	(B, 0.5)	(A, 0.6)	(B, 0.2)
3 [C]	(C, 0.1)	(C, 0.4)	(D, 0.7)
4 [A]	(D, 0.2)	(D, 0.3)	(C, 0.5)
5 [A]	(A, 0.3)	(C, 0.4)	(A, 0.5)
6 [B]	(B, 0.2)	(B, 0.3)	(D, 0.6)
7 [D]	(C, 0.4)	(C, 0.3)	(A, 0.5)

(a) Determine the predicted class for each test sample.

(b) Compute the confusion matrix for each class (A, B, C, D).

(c) Calculate the precision, recall, and F-1 score for each class.

(d) Compute the average accuracy.

(e) Compute the micro and macro F-1 scores.

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4. (6 points) Consider a multi-class classification problem with N classes and M test samples. Let TP, FP, and FN denote the total true positives, false positives, and false negatives when counts are aggregated over all classes (i.e., micro-averaging). Show that the micro-averaged F_1 score is equal to the overall accuracy, defined as $\text{Accuracy} = \frac{\# \text{ correct predictions}}{M}$. **Hint:** Relate the total number of predictions and the total number of actual labels to M , and observe how each misclassified sample simultaneously contributes to two error counts.

If a sample is FP, it is simultaneously a FN wrt its g.t label. Therefore, $\text{FP} = \text{FN} = M - \text{TP}$. Plug into $F_1 = \frac{2PR}{P+R}$.

5. (5 points) Consider a simple linear regression model of the form $\hat{y} = \beta_0 + \beta_1 x$ where x and y are the observed variables. Suppose both x and y have been standardized so that they each have mean zero and variance one. Prove that the estimated slope coefficient β_1 is equal to the sample correlation coefficient ρ_{xy} .

Model $\hat{y} = \beta_0 + \beta_1 x$ with standardized x, y : $\bar{x} = \bar{y} = 0$, $\text{Var}(x) = \text{Var}(y) = 1$.

OLS slope:

$$\beta_1 = \frac{\sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^N (x_i - \bar{x})^2} = \frac{\sum_{i=1}^N x_i y_i}{\sum_{i=1}^N x_i^2} = \frac{(N-1) s_{xy}}{(N-1) s_x^2} = \frac{s_{xy}}{s_x s_y} = \rho_{xy},$$

since $s_x = s_y = 1$.

Rough Work

